6.Implementation of Hierarchical Clustering

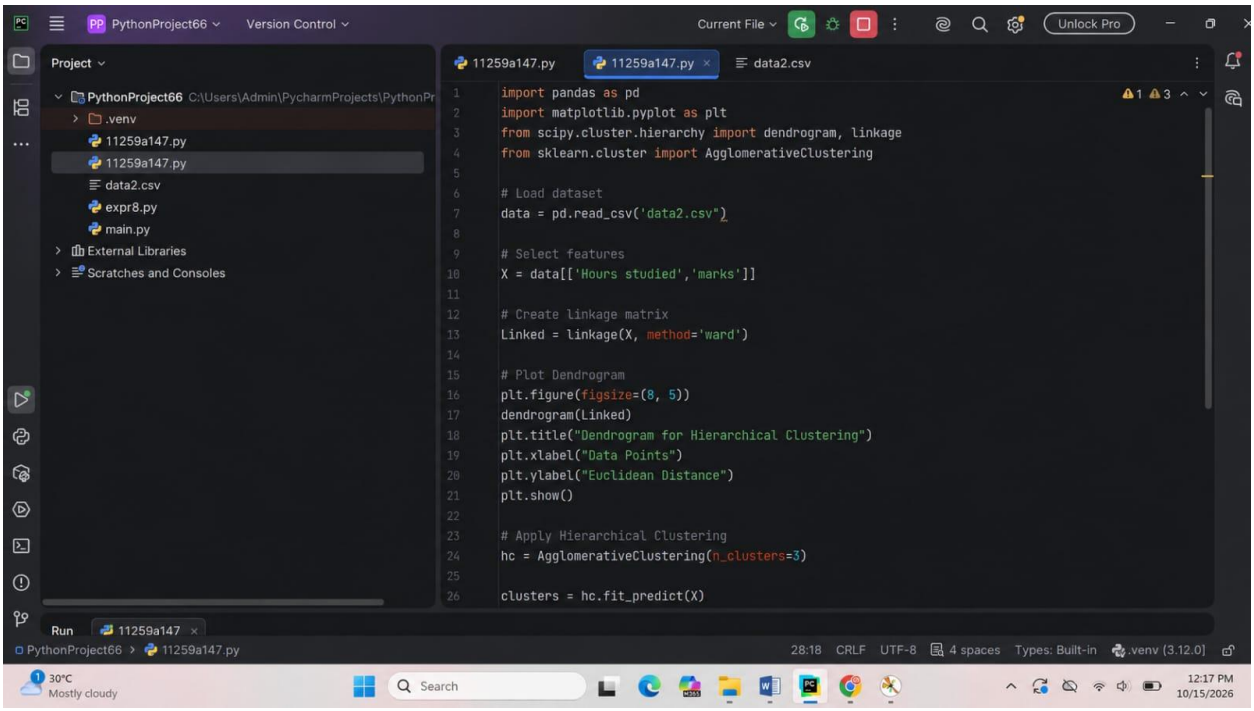
Aim:

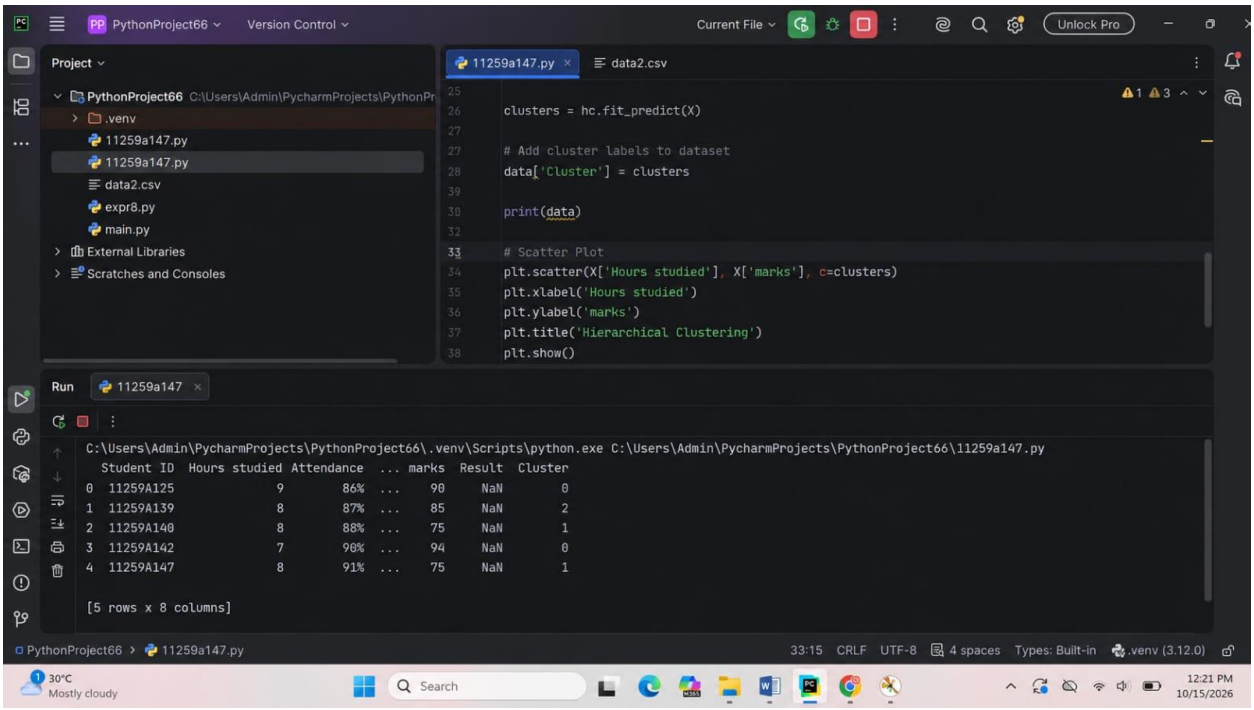
To write a program that implements hierarchical (agglomerative) clustering to group the given data points into clusters based on their similarity, and to display the cluster labels and a dendrogram showing how the clusters are merged.

Algorithm:

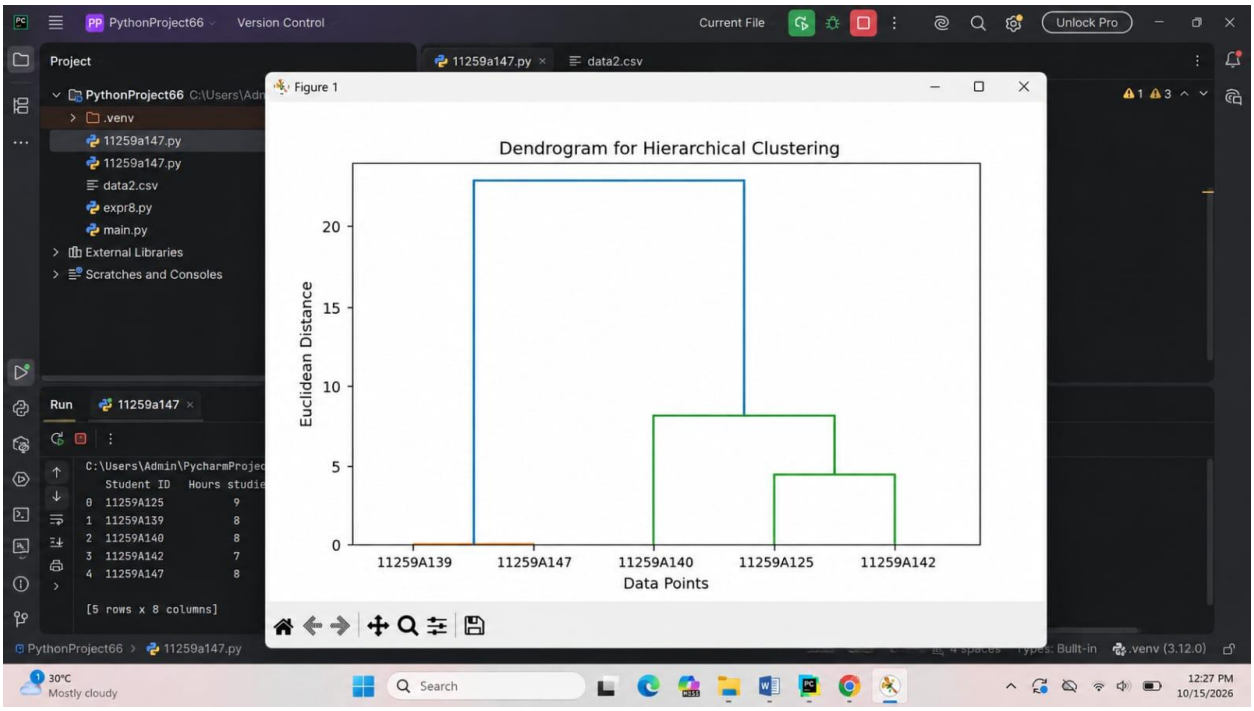
1. Start the program.
2. Import the required libraries (pandas, matplotlib, dendrogram and linkage from scipy.cluster.hierarchy, and AgglomerativeClustering from sklearn.cluster).
3. Load the dataset from the input file.
4. Select the input features to be used for clustering (no target variable is needed, since clustering is unsupervised).
5. Treat each data point as its own separate cluster.
6. Compute the distance between every pair of clusters (for example Euclidean distance with Ward linkage).
7. Merge the two closest clusters into a single cluster.
8. Update the distances between the new cluster and all the remaining clusters.
9. Repeat steps 7 and 8 until all the data points are merged into one cluster.
10. Plot the dendrogram to show the merging process and decide the number of clusters by cutting the tree at a suitable height.
11. Create the AgglomerativeClustering model with the chosen number of clusters (for example 3) and fit it on the data to obtain the cluster label for each data point.
12. Display the cluster labels and add them to the dataset.
13. Plot the data points coloured by cluster (optional).

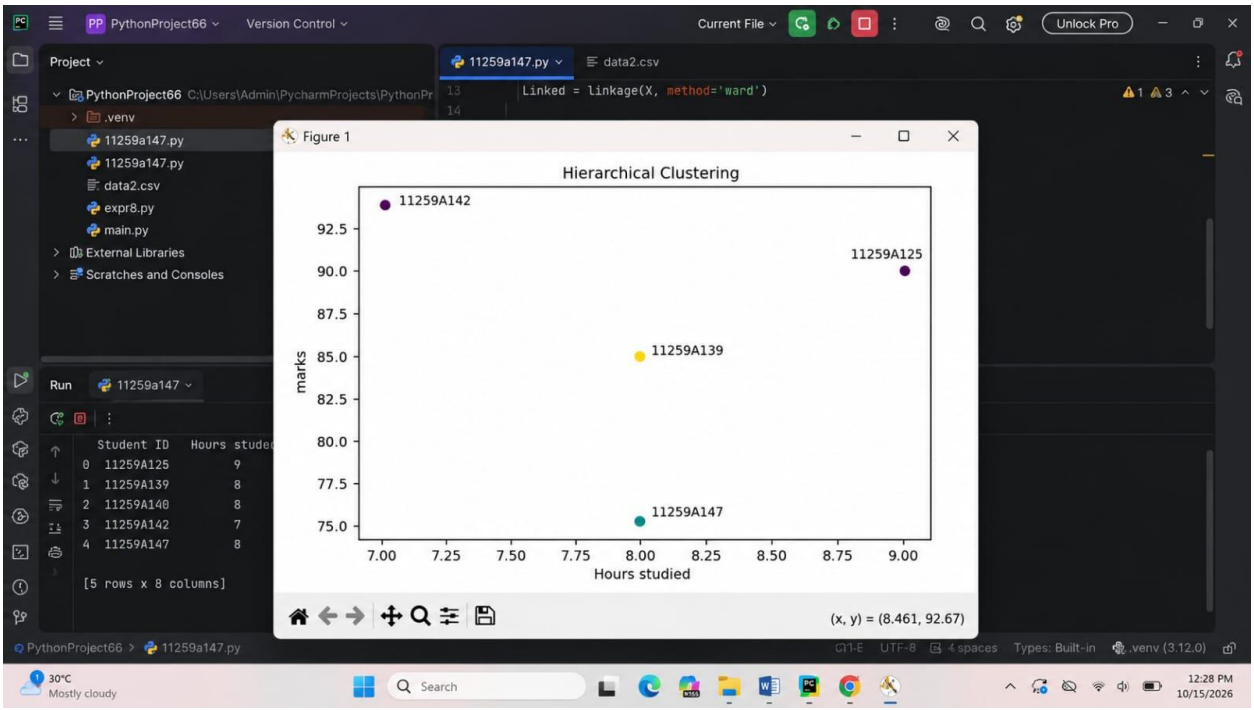
Program&Result:





Graph:


Figure 1
Dendrogram for Hierarchical Clustering
Euclidean Distance
0
5
10
15
20
11259A139
11259A147
11259A140
11259A125
11259A142
Data Points


Figure 1
Hierarchical Clustering
marks
92.5
90.0
87.5
85.0
82.5
80.0
77.5
75.0
11259A142
11259A125
11259A139
11259A147
7.00
7.25
7.50
7.75
8.00
8.25
8.50
8.75
9.00
Hours studied
(x, y) = (8.461, 92.67)